

# Jun-Hyung Jung, Ph.D.

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Busan, South Korea

## WORK EXPERIENCE

- **Pukyong National University** [\[🔗\]](#) Sept. 2026 - present  
Busan, South Korea  
*Division of Electrical Engineering, Major of Control and Instrument Engineering*
  - **Assistant Professor** in Power Electronics, Motor Drives, and Machine Learning
  - **Smart Power Electronics and Electric Drives (SPEED) Lab**
- **Fraunhofer Institute for Silicon Technology (ISIT)** [\[🔗\]](#) Jan. 2024 - Aug. 2026  
Kiel, Germany  
*Electronic Energy Systems Group*
  - **Associate Researcher:** Hybrid Grids and Power Converter Teams
  - **Project Leader:** 8 projects with industris and research institute (approx. total 900kEUR)
  - **AI representatives:** Manage AI-related issues and Define strategy for the institute
- **Chair of Power Electronics, Kiel University** [\[🔗\]](#) Oct. 2020 - Dec. 2023  
Kiel, Germany  
*Technical Faculty - Research Institute of Electrical Engineering and Information Technology*
  - **Postdoctoral Researcher:** Leading Postdoc for Multilevel Converter Research Group
  - Advisor: Prof. Dr.-Ing. Marco Liserre (IEEE Fellow)
- **Pusan National University** Oct. 2019 - Sept. 2020  
Busan, South Korea  
*South-East Grand ICT Research Center*
  - **Associate Researcher:** Development of grid-connected converters using WBG devices

## EDUCATION

- **Pusan National University** Mar. 2014 - Aug. 2019  
Busan, South Korea  
*Ph.D. in System and Controls, Department of Electrical and Computer Engineering*
  - Thesis: "Reduction of Common-Mode Voltage from Grid-Connected AC/DC Three-phase Three- level NPC PWM Converter System Using Fourth-Leg Active Filter."
  - Supervisor: Prof. Jang-Mok Kim
- **Pusan National University** Mar. 2012 - Feb. 2014  
Busan, South Korea  
*M.Sc. in Electrical Engineering, Department of Electrical and Computer Engineering*
  - Thesis: "A PWM Control Strategy for Low-speed Operation of Three-Level NPC Inverter Based on Bootstrap Gate Drive Circuit."
  - Supervisor: Prof. Jang-Mok Kim
- **Pusan National University** Mar. 2005 - Feb. 2012  
Busan, South Korea  
*B.Sc. in Electrical Engineering, Department of Electrical and Computer Engineering*

## TEACHING EXPERIENCE

- **Lecturer, "Microcontroller and FPGA Technique for Power Electronics Applications"** 2021 - 2025  
*M.Sc. Course, Kiel University*
  - Lecture and exercise course, 3 cred.h, ECTS studies, ECTS credits: 5
- **Teaching Assistant** 2020-2023  
*M.Sc. Course, Kiel University*
  - Lecture: Design of Power Electronics (2020-2022)
  - Practice: B.Sc. Practice (2021, 2023)

## RESEARCH INTEREST

- High-Power Power Conversion Systems with Parallel Operation of PWM Converters.
- Grid-connected AC-DC multilevel converters for medium voltage (MV) applications.
- MV Megawatt Solid-state Transformers for Modern Grid Systems
- MVDC applicable DC-DC converters for MW EV charging infrastructure and Energy Storage Systems.
- AI x Energy & Power: Reinforcement Learning for Energy and Power Electronics Applications.

## JOURNAL PUBLICATIONS

R= UNDER REVIEW, E= EARLY ACCESS ARTICLES, \* = CORRESPONDING AUTHOR

- [R1] J. Jung\*, S. Bazyar, D. D'amato, I. Jeong and M. Liserre "A Single-Carrier Rotating Modulation Strategy for Medium-Voltage Isolated Modular Multilevel DC-DC Converters," in IEEE Access, Under review.
- [E1] S. Bazyar, J. Jung, H. Beiranvand, H. Iman-Eini, and M. Liserre "Hybrid Time-Frequency Domain Modeling and Analysis of MMC-Based DC/DC Converters," in IEEE Transactions on Industrial Electronics, Early Access Articles.
- [1] Q. Xun, J. Jung\*, T. Wang, M. Liserre and D. Nowotka "A Two-Stage Framework for Optimal Battery Sizing and Reinforcement Learning based Energy Management of EV Charging Stations," in IEEE Transactions on Transportation Electrification, vol. 12, no. 2, pp. 2830-2842, April 2026. [📄]
- [2] R. Li, J. Jung\*, J. Diers, T. Hamann, H. Beiranvand, M. Liserre, Dirk Nowotka "Reinforcement Learning for Simultaneous SOC and Temperature Balancing of Li-Ion Battery Cells in Battery Management Systems," in IEEE Access, vol. 13, pp. 175357-175372, 2025. [📄]
- [3] R. Sancio, J. Jung\*, S. Pugliese, M. Langwasser and M. Liserre "Advanced Zero-Sequence Current Suppression in P-HIL Testbed with Integration of Feed-Forward Compensation and LADRC," in IEEE Access, vol. 12, pp.85400-85410, Jun. 2024. [📄]
- [4] J. Jung\*, S. Pugliese, and M. Liserre "Efficiency Enhanced Zero CM voltage Space Vector Modulation for Parallel Three-level Converters," in IEEE Transactions on Power Electronics, vol. 39, no 02, pp.1974-1979, Feb. 2024. [📄]
- [5] S. Brueske\*, S. Rupp, J. Jung, and M. Langwasser "Power-electronics-based distribution-flexible solutions tailored to the grid site," in e&I Elektrotechnik Informationstechnik, vol. 140, pp.634-644, Dec. 2023. [📄]
- [6] J. Jung\*, J. Kim, and M. Liserre "Conducted EMI Reduction by Active Power Filter Embedded in Neutral-Point-Clamped Converters," in IEEE Transactions on Power Electronics, vol. 38, no 12, pp.15551-15565, Dec. 2023. [📄]
- [7] S. Hwang, S. Cho, J. Jung, and J. Kim\*, "Phase Current Measurement Method of Dual Inverter-Motor Drive System Using a Single DC Link Current Sensor," in Energies, vol. 14, no.18, pp. 5626, Sept. 2021. [📄]
- [8] J. Jung, S. Hwang and J. Kim\*. "A Common-mode Voltage Reduction Method Using an Active Power Filter for a Three-Phase Three-Level NPC PWM Converter," in IEEE Transactions on Industry Applications, vol. 57, no. 4, pp. 3787-3800, Jul./ Aug. 2021. [📄]
- [9] H. Ku, J. Jung, J. Park, J. Kim\* and Y. Son. "Fault-Tolerant Control Strategy for Open-Circuit Fault of Two Parallel-Connected Three-Phase AC-DC Two-Level PWM Converter," in Journal of Power Electronics, vol. 20, no. 9, pp. 731-742, March. 2020. [📄]
- [10] J. Jung, H. Ku, W. Im and J. Kim\*, "A Carrier-based PWM Control Strategy for Three-level NPC Inverter Based on Bootstrap Gate Drive Circuit," in IEEE Transactions on Power Electronics, vol. 35, no 3, pp.2843-2860, March 2020. [📄]
- [11] J. Jung, H. Ku, Y. Son, and J. Kim\*, "Open-Switch Fault Diagnosis Algorithm and Tolerant Control Method of the Three-Phase Three-Level NPC Active Rectifier," in Energies, vol. 12, no.13, pp. 2495, Jun. 2019. [📄]
- [12] J. Jung, J. Park, J. Kim\* and Y. Son. "DC-Link Voltage Balance Control Using Fourth-Phase for 3-Phase 3-Level NPC PWM Converters with Common-Mode Voltage Reduction Technique," in Journal of Power Electronics, vol. 19, no. 1, pp. 108-118, Jan. 2019. [📄]
- [13] Y. Son, J. Jung and J. Kim\*, "Advanced field weakening control for squirrel-cage induction motor in wide range of DC-link voltage conditions" in Journal of Electrical Engineering and Technology, vol. 12, no. 2, pp. 665-673, Mar. 2017. [📄]

- [A1] M. Votava, H. Beiranvand, **J. Jung**, S. Bazyar, and M. Liserre, "Steady-State Model and Operating Analysis of an MMC-based Multiport DC-DC Converter", PCIM Europe 2026; International Exhibition and Conference for Power Electronics, Intelligent Motion, Renewable Energy and Energy Management, 2026.
- [A2] M. Votava, H. Beiranvand, **J. Jung**, S. Bazyar, and M. Liserre, "Steady-State Model and Operating Analysis of an MMC-based Multiport DC-DC Converter", PCIM Europe 2026; International Exhibition and Conference for Power Electronics, Intelligent Motion, Renewable Energy and Energy Management, 2026.
- [2] A. Eldistawy, **J. Jung**, E. D. Almwu, N. M. L. Tan, and M. Liserre, "Control of an Input-Series Output-Parallel Asymmetric Triple-Active-Bridge for Medium-Voltage DC Renewable Applications", in 2026 International Power Electronics Conference (IPEC), 2026.
- [3] **J. Jung**, D. D'amato, C. Park, and M. Liserre, "Thermal Stress Balancing Algorithm Using Nearest-Level PWM for Medium Voltage MMCs", in 2025 IEEE Energy Conversion Congress and Exposition Europe (ECCE Europe), 2025.
- [4] **J. Jung**, D. D'amato, Rui Li, and M. Liserre, "Voltage Balancing of Modular Multilevel DC-DC Converter for MV-LV DC Transformers", in 2025 IEEE Kiel PowerTech, 2025.
- [5] V. Y. Konda, **J. Jung**, Y. Pascal, and M. Liserre, "Junction Temperature Estimation in IGBT Modules using Machine Learning based Finetuning", PCIM Europe 2025; International Exhibition and Conference for Power Electronics, Intelligent Motion, Renewable Energy and Energy Management, 2025.
- [6] D. D'amato, **J. Jung**, and M. Liserre, "Discontinuous Modulation Strategy for Voltage and Temperature Balancing of MMCs", in 2025 IEEE Applied Power Electronics Conference and Exposition (APEC), 2025.
- [7] V. Y. Konda, **J. Jung**, and M. Liserre, "Intelligent Junction Temperature Estimation of an IGBT Using Machine Learning and Vce Measurement", in 2024 IEEE 15th International Symposium on Power Electronics for Distributed Generation Systems (PEDG), 2024.
- [8] S. Bazyar, **J. Jung**, H. Beiranvand, J. V. M. Farias, and M. Liserre, "Quasi Two-Level Modulation for the MMC-Based Isolated DC/DC Converter", in 2023 IEEE Energy Conversion Congress and Exposition (ECCE), 2023.
- [9] S. Bazyar, **J. Jung**, R. Li, H. Beiranvand, and M. Liserre, "Losses Analysis of MMC in Medium-Frequency Medium Voltage DAB Converter for Charging Station", in 2023 25th European Conference on Power Electronics and Applications (EPE'23 ECCE Europe), 2023.
- [10] K. Harwardt, **J. Jung**, H. Beiranvand, D. Nowotka, and M. Liserre, "Lithium-Ion Battery Management System with Reinforcement Learning for Balancing State of Charge and Cell Temperature", in 2023 IEEE Belgrade PowerTech, Belgrade, Serbia, 2023.
- [11] M. Hassanifar, V.M. Hrishikesan, **J. Jung**, S. Bazyar, H. Beiranvand, T. Pereira, M. Langwasser, and M. Liserre, "Modular Multilevel Converter Enabling Multibus DC Distribution", in 2023 IEEE 17th International Conference on Compatibility, Power Electronics and Power Engineering (CPE-POWERENG), 2023.
- [12] **J. Jung**, S. Bazyar, H. Beiranvand, J. V. M. Farias, and M. Liserre, "A Single Carrier Rotating Modulation for Modular Multilevel Converter based Isolated DC-DC Converters in EV Charging Station", 2023 11th International Conference on Power Electronics and ECCE Asia (ICPE 2023 - ECCE Asia), 2023.
- [13] **J. Jung**, E. Hosseini, M. Liserre and L. M. Fernández-Ramírez, "Reinforcement Learning Based Modulation for Balancing Capacitor Voltage and Thermal Stress to Enhance Current Capability of MMCs", in 2022 IEEE 13th International Symposium on Power Electronics for Distributed Generation (PEDG), 2022.
- [14] V. H. Nguyen, H. J. Hinrichs, **J. Jung** and M. Liserre, "Impedance Measurement of a Medium-Voltage Grid by a Mobile Measuring System with the Capability to Improve Power Quality", in 2022 IEEE 13th International Symposium on Power Electronics for Distributed Generation (PEDG), 2022.
- [15] **J. Jung**, S. Pugliese and M. Liserre, "Space Vector Modulation Technique for Reducing Harmonics in Current with Zero Common-Mode Voltage for Two-Parallel Three-Level Converters", IEEE Applied Power Electronics Conference and Exposition (APEC), 2022.
- [16] A. Tcai, T. Wijekoon, **J. Jung** and M. Liserre, "Quasi-Reference PWM for 3-level Voltage Source Inverters," 2021 IEEE Energy Conversion Congress and Exposition (ECCE), 2021.
- [17] **J. Jung**, M. Langwasser, S. Pugliese and M. Liserre, "Zero-sequence Circulating Current Suppression with Stand-alone Feedforward Control for Power Hardware-in-the-Loop System," 2021 IEEE Energy Conversion Congress and Exposition (ECCE), 2021.
- [18] **J. Jung**, S. Pugliese, M. Langwasser and M. Liserre, "Analysis and Suppression of Zero-Sequence Circulating Current in Multi-Parallel Converters," 2021 IEEE 15th International Conference on Compatibility, Power Electronics and Power Engineering (CPE-PowerEng), 2021.

- [1] **J. Jung**, N. Bisht, D. D'amato and C. Park, "Tutorial: Modular Multilevel Converters as a Backbone for Medium Voltage Grids", in 2025 IEEE Energy Conversion Congress and Exposition Europe (ECCE Europe), 2025.
- [2] M. Liserre, Y. Pascal, **J. Jung**, K. Debbadi, M. Votava and N. Bisht, "Seminar: High-Power Converters: Active Reliability & Fault-Tolerant Operation", PCIM Europe 2024; International Exhibition and Conference for Power Electronics, Intelligent Motion, Renewable Energy and Energy Management, 2024.

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## SUPERVISION OF THESIS WORKS

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- [1] Mohammad Ashik Amin, M.Sc. Degree Thesis at Kiel University, Germany "Fault Diagnosis Algorithm for Modular Multilevel Isolated Dual-Active-Bridge DC-DC Converter", in 2026.
- [2] Stayner Nobrega Barros, M.Sc. Degree Thesis at Kiel University, Germany "Active Thermal Control with DPWM for Modular Multilevel Converters", in 2024.
- [3] Mohamed Eissa, M.Sc. Degree Thesis at École Centrale de Nantes, France "Machine Learning Based Estimation of Sub-module Status in Modular Multilevel Converter", in 2024.
- [4] Dominik Frymark, M.Sc. Degree Thesis at Kiel University, Germany "Machine Learning based Power Management for Solar-powered Residential Building Applications", in 2023.
- [5] Tailei Wang, M.Sc. Degree Thesis at Kiel University, Germany "Optimal Design and Machine Learning based Power Management for an EV Charging Station with Energy Storage System", in 2023.
- [6] Essam Arifi, M.Sc. Degree Thesis at Kiel University, Germany "Intelligent Fault Diagnosis for Modular Multilevel Converters", in 2023.
- [7] Mohammad El-Iraky, M.Sc. Degree Thesis at Kiel University, Germany "Kalman-Filter based Thermal Digital Twin", in 2023.
- [8] Rui Li, M.Sc. Degree Thesis at Kiel University, Germany "Power Loss Analysis of Medium-Voltage Isolated DC-DC Converter for EV Charging Station", in 2022.
- [9] Andreas Zahner, B.Sc. degree thesis at Kiel University, Germany "Machine Learning based Control for Modular Multilevel Converters", in 2022.